

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 13-2026: A 76-Year-Old Woman with Fatigue, Rash, and Kidney Failure

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PRESENTATION OF CASE

Dr. Tushar Kamath (Medicine): A 76-year-old woman was transferred to the medical service of this hospital because of rapidly progressive kidney failure.

The patient had been in her usual state of health until 1 year before the current presentation, when a painless purpuric rash on the lower legs developed. The rash waxed and waned. There was no joint pain, edema, weakness, or weight loss. The patient was seen by a dermatologist at another hospital, and the results of a skin biopsy were reportedly consistent with leukocytoclastic vasculitis. The results of urinalysis and kidney-function tests were normal. Her symptoms were attributed to a history of hepatitis C virus (HCV) infection; topical glucocorticoid treatment was initiated, and the rash resolved.

Four weeks before the current presentation, the patient had persistent sinonasal congestion, drainage, and pain, and she received amoxicillin as treatment for presumptive bacterial sinusitis. The sinonasal symptoms resolved, but after she completed the course of amoxicillin, the purpuric rash on the lower legs recurred and was accompanied by arthralgias, ankle edema, and fatigue. The patient was evaluated in the rheumatology clinic of the other hospital. Physical examination was notable for pitting edema extending from the ankles to the shins and palpable purpura on the lower legs and thighs, without synovitis or motor weakness. Glucocorticoid treatment was deferred, and additional laboratory evaluation was planned.

Two weeks before the current presentation, headache, confusion, and difficulty with word finding developed, prompting the patient to present to the emergency department of the other hospital. The blood pressure, heart rate, and temperature were normal. Physical examination revealed a mild facial droop on the right side, without pronator drift or upper-arm weakness. Computed tomographic angiography of the head and neck, performed after the administration of intravenous contrast material, was reportedly negative for infarction, large-vessel occlusion, and high-grade stenosis. Magnetic resonance imaging (MRI) of the head reportedly revealed a small subacute infarct in the left posterior insula. Treatment with aspirin and rosuvastatin was started.

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CME



The patient's hospital course was also notable for the development of acute kidney injury, with elevated serum creatinine levels as high as 2.6 mg per deciliter (230 μ mol per liter; reference range, 0.6 to 1.5 mg per deciliter [53 to 133 μ mol per liter]), as compared with a previously reported baseline level of 0.8 mg per deciliter (71 μ mol per liter). The elevation in serum creatinine levels was attributed to the administration of intravenous contrast material. Additional laboratory test results are shown in Table 1. A follow-up visit in the nephrology clinic was scheduled, and the patient was discharged home.

Three days before the current presentation, the patient underwent additional laboratory testing at the nephrology clinic of the other hospital. The results showed progressive kidney dysfunction, with a serum creatinine level of 4.2 mg per deciliter (371 μ mol per liter), as well as a C3 level of 73 mg per deciliter (reference range, 81 to 157) and a C4 level of less than 6 mg per deciliter (reference range, 12 to 39). Additional laboratory test results are shown in Table 1. Urinalysis showed 3+ blood (reference value, negative), more than 180 red cells per high-power field (reference range, 0 to 2), and 3+ protein (reference value, negative). Quantification of protein in a 24-hour urine collection showed 4.4 g of total protein per day.

Given the rapidly progressive kidney failure, the patient was instructed to present to the emergency department of the other hospital. At the other hospital, ultrasonography of the kidneys and bladder reportedly showed no evidence of hydronephrosis. In the context of progressive oliguria, hypervolemia, and hyperkalemia, a temporary hemodialysis catheter was placed, and hemodialysis was started. Treatment with high-dose intravenous methylprednisolone was initiated, and the patient was transferred to this hospital.

On presentation to this hospital, the patient reported a headache but felt well otherwise. The review of symptoms was negative for Raynaud's phenomenon, numbness or tingling, epistaxis, nasal crusting, focal weakness, weight loss, night sweats, and rigors. The medical history was notable for HCV infection, which the patient had acquired more than 30 years earlier from a blood transfusion in the context of traumatic hemorrhage. Fifteen years earlier, HCV infection had been treated with sofosbuvir, which had led to a sustained virologic cure. The patient also had a

history of hyperlipidemia, major depressive disorder, recurrent urinary tract infections, and hypothyroidism. The surgical history was notable for cholecystectomy. Medications included levothyroxine, rosuvastatin, aspirin, and citalopram. There were no known adverse reactions to medications.

The patient was a retired administrator, and she resided with her husband in an assisted-living facility in a suburban area of New England. She did not smoke, drink alcohol, or use cocaine or other recreational drugs. She had no recent travel and had not lived outside the United States. The family history was negative for kidney disease; her mother had had ovarian cancer.

On examination, the oral temperature was 36.7°C, the blood pressure 138/65 mm Hg, the heart rate 85 beats per minute, the respiratory rate 16 breaths per minute, and the oxygen saturation 97% while the patient was breathing ambient air. The weight was 88.8 kg, and the body-mass index (the weight in kilograms divided by the square of the height in meters) was 38. Pitting edema extending from the ankles to the thighs was noted, along with a faint high-amplitude tremor in the arms. Mild jugular venous distention was present, and there was a hemodialysis catheter in the neck. Cardiac auscultation revealed a blowing holosystolic murmur at the left lower sternal border. Fading hyperpigmented macules were visible throughout the lower legs.

HCV RNA was not detected in the blood. Screening tests for human immunodeficiency virus and hepatitis B virus were negative. Antinuclear antibodies, antineutrophil cytoplasmic antibodies (ANCA), and antibodies against Ro and La were not detected. Anti-cyclic citrullinated peptide antibodies were not detected, but the rheumatoid factor level was elevated at 46 IU per milliliter (reference range, 0 to 14). The complete blood count was notable for a white-cell count of 14,690 (reference range, 4000 to 11,000), with neutrophil predominance. Blood was obtained for culture. Additional laboratory test results are shown in Table 1.

Dr. Otto Rapalino: MRI of the head was performed on admission. Moderate, ill-defined areas of abnormal hyperintensity in the supratentorial white matter were seen on fluid-attenuated inversion recovery imaging, a finding compatible with chronic small-vessel disease (Fig. 1A). A punctate focus of acute or early subacute infarction

Table 1. Laboratory Data.*

Variable	Reference Range, Adults, This Hospital†	2 Wk before Current Presentation, Other Hospital	3 Days before Current Presentation, Other Hospital	On Current Presentation, This Hospital
Hemoglobin (g/dl)	12.0–16.0	—	8.4	9.6
Hematocrit (%)	41.0–53.0	—	26.4	29.2
White-cell count (per μ l)	4500–11,000	—	5600	14,690
Differential count (per μ l)				
Neutrophils	1800–7700	—	4370	13,530
Lymphocytes	1000–4800	—	680	700
Monocytes	200–1200	—	380	310
Eosinophils	0–900	—	100	0
Basophils	0–300	—	30	10
Immature granulocytes	0–100	—	0	140
Platelet count (per μ l)	150,000–400,000	—	284	260
Sodium (mmol/liter)	135–145	139	137	137
Potassium (mmol/liter)	3.4–5.0	4.3	4.8	4.4
Chloride (mmol/liter)	98–108	108	100	103
Carbon dioxide (mmol/liter)	23–32	20	20	25
Urea nitrogen (mg/dl)	8–25	39	49	34
Creatinine (mg/dl)	0.6–1.5	2.6	4.2	2.6
Calcium (mg/dl)	8.5–10.5	9.2	9.0	8.8
Glucose (mg/dl)	70–110	99	100	103
Aspartate aminotransferase (U/liter)	9–32	—	—	25
Alanine aminotransferase (U/liter)	7–33	—	—	17
Alkaline phosphatase (U/liter)	30–100	—	—	68
Total bilirubin (mg/dl)	0.0–1.0	—	—	0.2
Albumin (g/dl)	3.3–5.0	—	—	3.0
Globulin (g/dl)	1.9–4.1	—	—	2.4
C3 (mg/dl)	81–157	—	73	41
C4 (mg/dl)	12–39	—	<6	<6
IgG (mg/dl)	614–1295	—	—	78
IgA (mg/dl)	69–309	—	—	109
IgM (mg/dl)	53–334	—	—	466
Free kappa light chain (mg/liter)	3.3–19.4	—	—	356.3
Free lambda light chain (mg/liter)	5.7–26.3	—	—	18.1
Kappa:lambda ratio	0.30–1.70	—	—	19.69
Serum protein electrophoresis	Negative for monoclonal component	—	—	IgM kappa M component detected (0.12 g/dl)

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for bilirubin to micromoles per liter, multiply by 17.1.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

was present in the left parieto-occipital parenchyma (Fig. 1B), along with a few nonspecific parenchymal microhemorrhages (Fig. 1C). A punctate focus of enhancement in the right putamen without restricted diffusion was suggestive of late subacute infarction (Fig. 1D). Three-dimensional time-of-flight magnetic resonance angiography (MRA), performed without the ad-

ministration of contrast material, revealed focal critical stenosis or focal occlusion of the middle portion of the M2 segment of the left middle cerebral artery, along with focal stenoses of the proximal portion of the M2 segment of the left middle cerebral artery and the P3 segment of the left posterior cerebral artery (Fig. 1E and 1F). No concentric enhancement of the vessel wall was

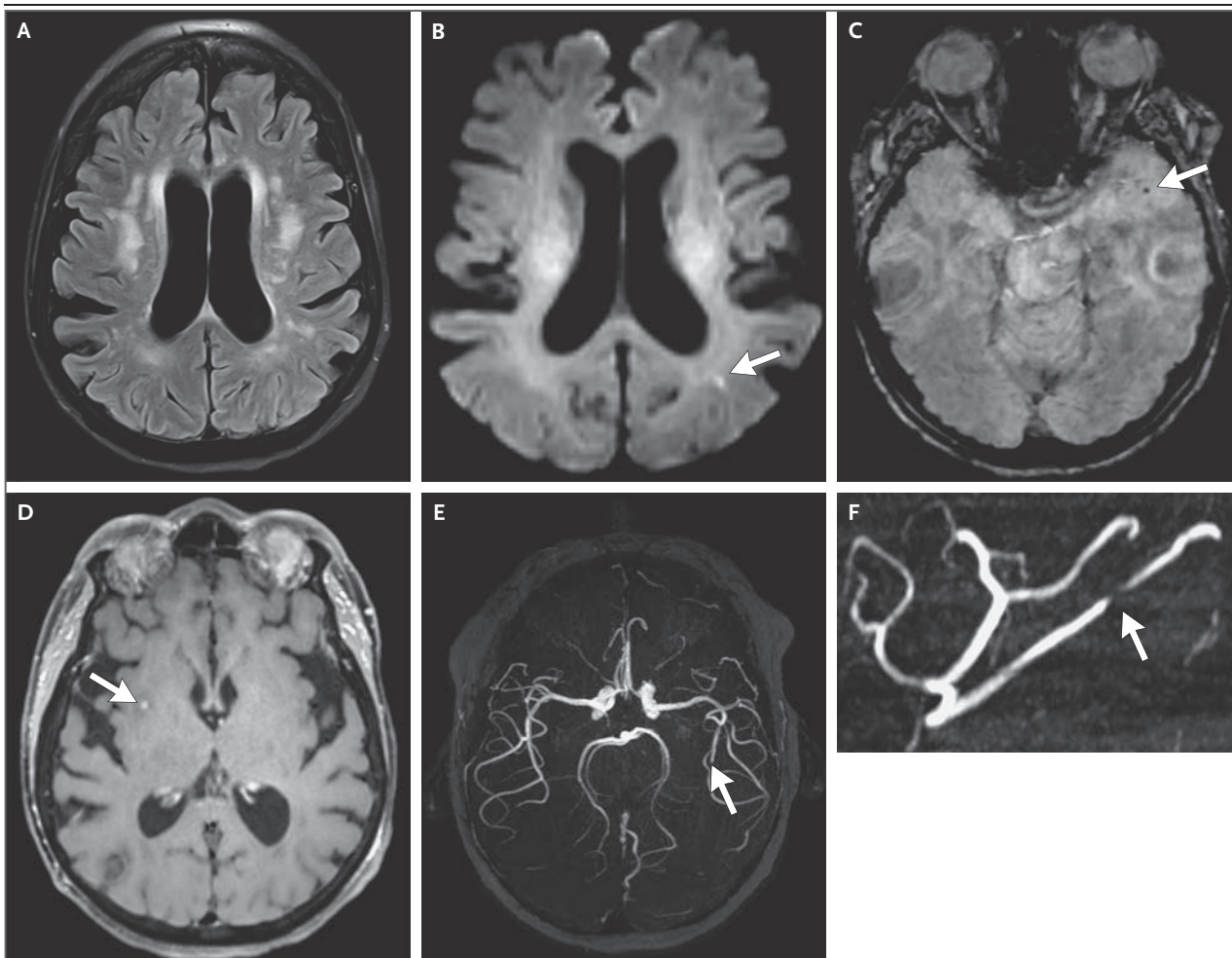


Figure 1. MRI and MRA of the Head.

MRI of the head was performed on admission. An axial fluid-attenuated inversion recovery image (Panel A) shows moderate, nonspecific hyperintensities that are compatible with chronic small-vessel disease. An axial diffusion-weighted image (Panel B) shows a punctate focus of acute or early subacute infarction in the left parieto-occipital parenchyma (arrow). An axial susceptibility-weighted image (Panel C) shows a nonspecific microhemorrhage in the left temporal lobe (arrow). An axial contrast-enhanced T1-weighted black-blood image (Panel D) shows a punctate focus of enhancement in the right putamen (arrow) without corresponding restricted diffusion, a finding suggestive of late subacute infarction. No abnormal vascular-wall enhancement is shown in the visualized arteries. Magnetic resonance angiography (MRA) was also performed. Axial and sagittal images (Panels E and F, respectively) show focal critical stenosis or occlusion of the middle portion of the M2 segment of the left middle cerebral artery (arrows), along with focal stenoses of the proximal portion of the M2 segment of the left middle cerebral artery and the P3 segment of the left posterior cerebral artery.

seen at the locations of the abnormalities. MRA of the neck was degraded by artifacts but showed no high-grade stenosis of the cervical arteries.

Dr. Kamath: A diagnosis was made.

DIFFERENTIAL DIAGNOSIS

Dr. Frank B. Cortazar: This 76-year-old woman with a history of treated HCV infection first reported a purpuric rash 1 year before the current presentation. A skin biopsy revealed leukocytoclastic vasculitis, a histopathological finding that is commonly seen with most types of cutaneous small-vessel vasculitis. At that time, there was no evidence of extracutaneous involvement, and the purpura resolved with the administration of topical glucocorticoids.

Within a month before the current presentation, the patient had recurrent purpura that was accompanied by arthralgias, proteinuria, an active urinary sediment, and deteriorating kidney function. The urinary profile, coupled with the rapid decline in kidney function, is consistent with the clinical syndrome of rapidly progressive glomerulonephritis (RPGN).¹ This patient's clinical problem can best be represented as a systemic small-vessel vasculitis with severe kidney involvement.

CONCEPTUAL FRAMEWORK FOR SYSTEMIC SMALL-VESSEL VASCULITIS

The pathogenesis of small-vessel vasculitis with glomerular involvement can be conceptualized in terms of the three immunofluorescence patterns that can be seen on kidney biopsy: linear, pauci-immune, and immune-complex. The linear pattern is found in patients with anti-glomerular basement membrane (anti-GBM) disease that is due to autoantibodies directed against the NC1 (noncollagenous 1) domain of the alpha 3 chain of type IV collagen.² The pauci-immune pattern, which is characterized by minimal immunoglobulin and complement deposition, is found in patients with ANCA-associated vasculitis.³ The immune-complex pattern, with granular immune-complex deposition, is observed in those with other types of small-vessel vasculitis, including IgA vasculitis, cryoglobulinemic vasculitis, systemic lupus erythematosus, and infection-associated glomerulonephritis.

In formulating an initial differential diagnosis, consideration of disease prevalence can be helpful. Among patients presenting with RPGN, the most likely cause is heavily influenced by age.⁴ Immune-complex diseases predominantly affect younger patients, whereas ANCA-associated vasculitis is more likely to occur in older patients. In one analysis, approximately 80% of patients presenting with RPGN who were older than 60 years of age had ANCA-associated vasculitis.⁴

Before additional data are considered, this patient presenting with small-vessel vasculitis and RPGN has a high pretest probability of ANCA-associated vasculitis. Anti-GBM disease targets only the glomerular and alveolar basement membranes, and this disease alone cannot explain the cutaneous vasculitis seen in this patient. However, approximately one third of patients with anti-GBM disease also have ANCA-associated vasculitis, and therefore, "double positivity" is a diagnostic consideration.⁴ Assessment of the available laboratory data can be helpful in further refining the differential diagnosis.

SERUM COMPLEMENT LEVELS IN THE EVALUATION OF GLOMERULONEPHRITIS

In a patient with glomerulonephritis, obtaining serum levels of C3 and C4 is a useful first step.¹ This patient has an undetectable C4 level and a mildly low C3 level. Given that patients with ANCA-associated vasculitis, anti-GBM disease, or IgA vasculitis typically present with normal serum complement levels, these diagnoses can be ruled out on the basis of the patient's complement profile.

In patients with hypocomplementemia, the relative reduction in C3 and C4 levels can provide insight into disease pathogenesis, suggesting specific causes.⁵ Diseases that predominantly activate the alternative complement pathway, such as infection-associated glomerulonephritis, typically cause a preferential decrease in the C3 level. Diseases that activate the classical complement pathway, such as cryoglobulinemic vasculitis and systemic lupus erythematosus, often lead to suppression of both C3 and C4. Cryoglobulinemic vasculitis is associated with a disproportionately low C4 level.⁶ New onset of systemic lupus erythematosus would be an unusual diagnosis in this patient's age group, and a negative test for anti-

nuclear antibodies rules out the diagnosis. The clinical picture of cutaneous vasculitis and RPGN, coupled with an undetectable C4 level, suggests the possibility of cryoglobulinemic vasculitis.

CRYOGLOBULINEMIC VASCULITIS

Cryoglobulins are immunoglobulins that precipitate from serum in vitro at temperatures below 37°C and redissolve upon rewarming.⁶ When cryoglobulins are detected, their composition should be determined to guide additional evaluation. Type I cryoglobulins consist of a single monoclonal immunoglobulin, typically IgG or IgM. Type II and type III cryoglobulins are mixed cryoglobulins that usually consist of IgM with rheumatoid factor activity and polyclonal IgG. In type II cryoglobulins, the IgM component is monoclonal (typically IgM kappa), whereas in type III cryoglobulins, the IgM component is polyclonal.

This patient had a positive test for rheumatoid factor and had a spike in IgM kappa, findings that are most consistent with type II cryoglobulins. Whereas type I cryoglobulins typically cause tissue injury through precipitation in small vessels, which leads to mechanical obstruction, mixed cryoglobulins are potent activators of the classical complement cascade and cause overt vasculitis.⁷ This patient has documented cutaneous vasculitis and would be expected to have a membranoproliferative pattern of injury on kidney biopsy, which is the histopathological finding typically observed with cryoglobulinemic glomerulonephritis.

CAUSES OF TYPE II CRYOGLOBULINEMIC VASCULITIS

The monoclonal component of type II cryoglobulins is indicative of a clonal population. Chronic HCV infection has historically been the main cause of type II cryoglobulinemia.⁷ Interaction between the E2 protein of HCV and a membrane complex on B lymphocytes leads to lymphoproliferation, which can progress to clonality and, in some cases, overt B-cell lymphoma.⁸ In one retrospective series, patients with HCV infection and type II cryoglobulins had a risk of lymphoma 35 times that seen in the general population.⁹ This patient has a history of HCV infection that was treated with sofosbuvir 15 years before the current presentation, which led to a sustained

virologic cure. She therefore does not have classic HCV infection–associated cryoglobulinemic vasculitis. It is possible, albeit unlikely, that her previous HCV infection initiated a clonal process, which has now become clinically apparent.

Autoimmune diseases, particularly Sjögren's syndrome, are also associated with the development of type II cryoglobulins. Like HCV infection, Sjögren's syndrome can initiate a clonal process, which can lead to type II cryoglobulinemia and lymphoma.¹⁰ The absence of sicca symptoms and the negative testing for Ro and La antibodies are features that rule out a diagnosis of Sjögren's syndrome in this patient.

In a case series involving patients with type II cryoglobulinemic vasculitis that excluded those with HCV infection, 91% of the patients had a documented B-cell clonal process.¹¹ Approximately one third of these patients met the criteria for an overt hematologic disorder, such as marginal-zone lymphoma or lymphoplasmacytic lymphoma. The remaining patients had a monoclonal gammopathy of renal significance. In either scenario, clone-directed therapy is indicated for treatment of the resulting vasculitis.

REMAINING QUESTIONS

In solving complex clinical problems, a useful exercise is to review features of the case that are not readily explained by the presumptive diagnosis. Habituation of this process can reduce diagnostic errors and lead to the recognition of concomitant disease processes. In this case, the stroke, cardiac murmur, and hypogammaglobulinemia merit further consideration.

The patient has a lacunar infarct. Of note, she does not have a history of hypertension, the main risk factor for lacunar infarcts.¹² Because the infarct developed in the context of clinically significant vasculitic activity, a rare occurrence of cryoglobulinemic vasculitis with involvement of the cerebral vasculature should be considered. Given the newly identified cardiac murmur, a cardioembolic process should also be considered as the underlying cause of stroke, although such processes are not often associated with lacunar infarcts.

The patient has a holosystolic murmur at the left sternal border, a finding that is most often associated with tricuspid regurgitation but also

occurs in some cases of mitral regurgitation. The murmur, coupled with vasculitis and a recent stroke that could be embolic, is suggestive of endocarditis. Infective endocarditis can lead to cutaneous vasculitis, immune-complex glomerulonephritis, and rheumatoid factor positivity.¹³ However, the extended course of this patient's illness, with cutaneous vasculitis developing 1 year earlier, rules out endocarditis as the sole driver of the patient's presentation. Moreover, the rheumatoid factor is polyclonal in patients with endocarditis, whereas in this patient, the rheumatoid factor is most likely the IgM kappa paraprotein. Nevertheless, given the overlapping manifestations and the consequences of a missed diagnosis, blood cultures and echocardiography should be performed before immunosuppressive therapy is administered.

The patient also has a low IgG level, a finding that is not typical of cryoglobulinemic vasculitis. Formal quantification of urinary protein shows nephrotic-range proteinuria. In combination with hypoalbuminemia, this finding indicates the presence of nephrotic syndrome, which can cause hypogammaglobulinemia due to urinary loss of immunoglobulins.¹⁴ This patient also has a high probability of having a lymphoproliferative disorder, which can lead to hypogammaglobulinemia through a variety of mechanisms.¹⁵ If the patient's hypogammaglobulinemia cannot be explained by either of these processes, additional workup to identify the underlying cause is indicated.

To confirm the diagnosis of type II cryoglobulinemic vasculitis in this patient, serum cryoglobulin levels should be obtained and a kidney biopsy should be performed. Once the diagnosis is confirmed, a bone marrow biopsy should be performed to evaluate for a lymphoproliferative disorder.

CLINICAL IMPRESSION

Dr. Meghan E. Sise: This 76-year-old woman, who had no history of kidney disease, presented with RPGN. The patient's serologic findings and purpuric rash strongly suggested cryoglobulinemic vasculitis. She had a remote history of treated HCV infection, and we had a high suspicion that her cryoglobulinemia was driven by a lymphoproliferative disorder.

HCV infection is associated with an increased risk of B-cell non-Hodgkin lymphoma and other lymphoproliferative disorders¹⁶; the risk decreases with a sustained virologic response¹⁷ but may not be fully eliminated in all patients. Even after HCV infection is cured, B-cell clones may persist, and thus, the risk of lymphoproliferative disorders may remain in some patients.¹⁸

In this patient, a kidney biopsy and serum cryoglobulin testing had been pursued at the other hospital. To address the life-threatening manifestations of cryoglobulinemic vasculitis, including the RPGN and neurologic involvement, we planned to perform urgent plasma exchange and to continue treatment with methylprednisolone. To determine the definitive immunosuppressive regimen, we needed to rule out cancer, and the oncology team planned an urgent bone marrow biopsy. Unfortunately, on the day that the patient was transferred to this hospital, rigors and fever developed, and blood cultures revealed high-grade *Staphylococcus aureus* bacteremia in four of four bottles. As a result, we withheld plasma exchange and performed an urgent replacement of her temporary dialysis catheter.

CLINICAL DIAGNOSIS

Cryoglobulinemic glomerulonephritis due to a lymphoproliferative disorder.

DR. FRANK B. CORTAZAR'S DIAGNOSIS

Cryoglobulinemic glomerulonephritis due to a lymphoproliferative disorder.

PATHOLOGICAL DISCUSSION

Dr. Claire Trivin-Avillach: The kidney biopsy revealed a membranoproliferative pattern of injury, with diffuse endocapillary hypercellularity and few cellular crescents (Fig. 2A). Many intracapillary loops had pseudothrombi (Fig. 2B). Immunofluorescence staining showed numerous deposits along the glomerular basement membranes and in the capillary lumens that were predominantly reactive for IgM and kappa (Fig. 2C through 2F). Abundant intracapillary, subendothelial, and

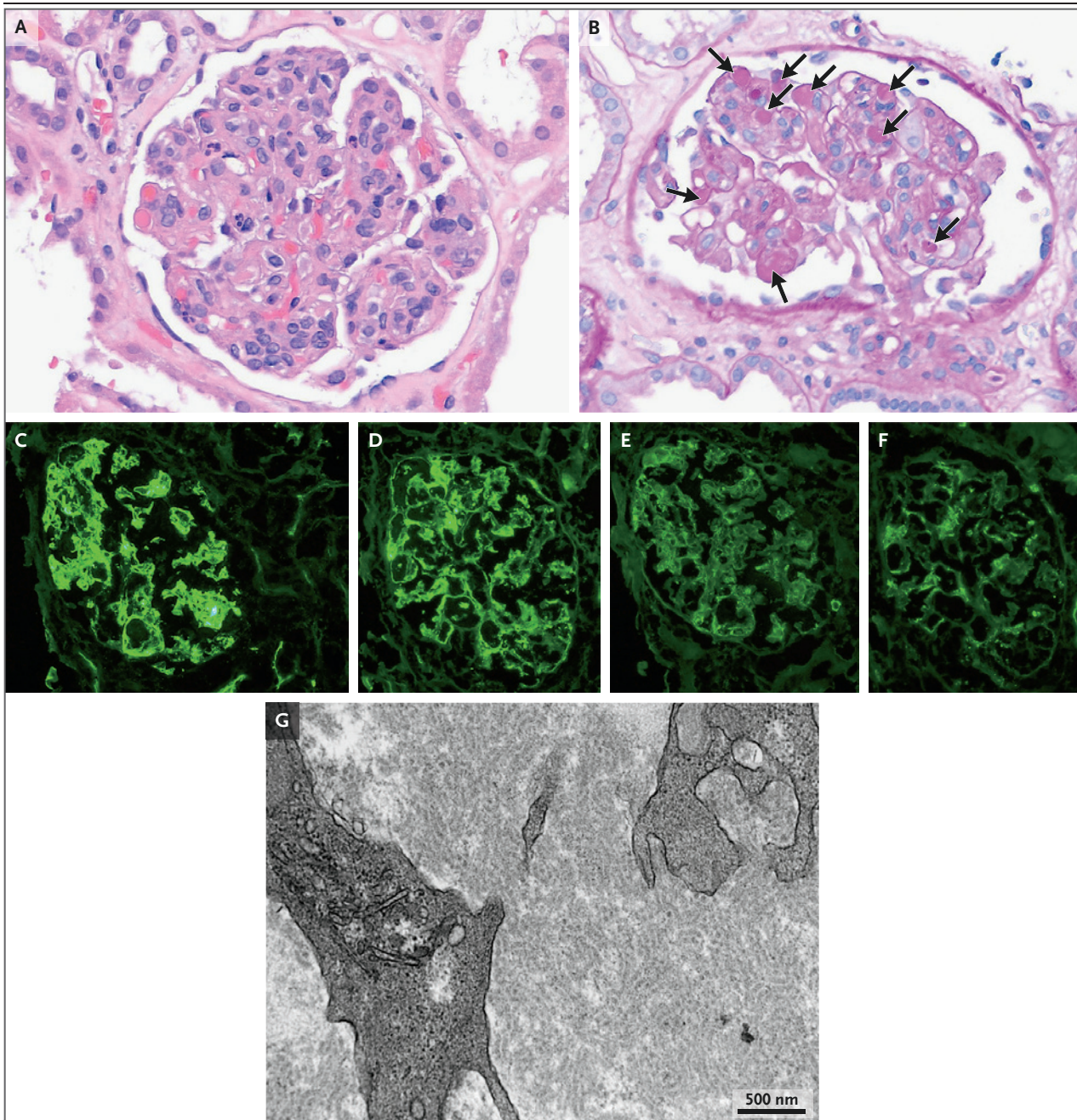


Figure 2. Kidney-Biopsy Specimens.

Hematoxylin and eosin staining (Panel A) shows a glomerulus with a membranoproliferative pattern of injury, including endocapillary hypercellularity with polymorphonuclear neutrophils and intracapillary eosinophilic material that is suggestive of pseudothrombi. Periodic acid–Schiff staining (Panel B) is positive in intracapillary material (arrows), a finding typical of pseudothrombi; negative staining would be typical of fibrin thrombi. Immunofluorescence staining (Panels C through F) shows deposits along the glomerular basement membranes, in the mesangium, and in the capillary lumens that are reactive for IgM (3+) and kappa (3+) (Panels C and D, respectively) and for C3 (2+) (not shown), with lesser reactivity for IgG (1+) and lambda (1+) (Panels E and F, respectively) and with no reactivity for C1q and IgA (not shown). Electron microscopy (Panel G) shows mesangial deposits with microtubular organization and a curvilinear pattern.

mesangial electron-dense deposits with tubular, curvilinear substructures were identified ultrastructurally (Fig. 2G). These features are typical of cryoglobulinemic glomerulonephritis. The pattern of injury, with marked glomerulitis, and the immunofluorescence pattern favor a mixed cryoglobulinemia; however, the dominance of IgM and kappa suggests the possibility of type I (monoclonal) cryoglobulins. Serum test results were consistent with type II (mixed) cryoglobulins with a dominant IgM kappa component.

A bone marrow core biopsy revealed a moderately hypercellular marrow characterized by maturing trilineage hematopoiesis with myeloid predominance and rare nonparatrabeular lymphoid aggregates. These aggregates were composed mainly of B cells with few plasma cells (Fig. 3A and 3B) and predominantly expressed IgM (Fig. 3C) and kappa (Fig. 3D and 3E). Flow cytometry of the bone marrow aspirate identified a small population of B cells (CD19+ and CD10–) with an excess of kappa light chains (kappa:lambda ratio in bone marrow aspirate, 3.6; reference range, 1.0 to 3.0). Plasma cells accounted for 8% of the total cellularity and had kappa predominance (kappa:lambda ratio, 4.4). Flow cytometry of the peripheral blood revealed a concordant population of B cells with monotypic kappa expression (Fig. 3F). These findings are consistent with low-level involvement by a small mature B-cell lymphoma with plasmacytic differentiation — either a marginal-zone lymphoma or lymphoplasmacytic lymphoma.¹⁹ The MYD88 c.794T→C (p.L265P) mutation, which is present in the vast majority of cases of lymphoplasmacytic lymphoma, was not detected.

PATHOLOGICAL DIAGNOSIS

Type II cryoglobulinemic vasculitis associated with a small mature B-cell lymphoma with plasmacytic differentiation.

DISCUSSION OF MANAGEMENT

Dr. Andrew R. Branagan: Symptomatic cryoglobulinemia — whether monoclonal type I or mixed type II — is an indication for the initiation of therapy for Waldenström macroglobulinemia.^{20,21} Waldenström macroglobulinemia, the most com-

mon subtype of lymphoplasmacytic lymphoma, is characterized by the presence of detectable IgM monoclonal protein. Although the MYD88 L265P mutation is found in more than 90% of patients with Waldenström macroglobulinemia, this patient appeared to have a wild-type mutation. False negative results of genetic sequencing may occur, particularly when testing is performed on unsorted bone marrow specimens with a low tumor burden.²² MYD88–wild-type Waldenström macroglobulinemia is associated with more aggressive clinical features and worse outcomes than MYD88-mutated disease.

Initial therapy with a proteasome inhibitor–based regimen that consisted of bortezomib, dexamethasone, and rituximab was selected because of its activity in patients with kidney dysfunction. Proteasome inhibitor–based regimens are also effective across genomic subgroups of Waldenström macroglobulinemia.²³ Shortly after the initiation of therapy, the patient had an acute transient increase in IgM levels with worsening cryoglobulinemic manifestations, a response consistent with a rituximab-associated IgM flare.

It is important to note that treatment involved multiple approaches. Therapeutic plasma exchange was initiated to rapidly reduce circulating IgM and cryoglobulin levels until clone-directed therapy took effect. Additional supportive measures included avoidance of cold exposure, warming of blood products and intravenous fluids, and the administration of systemic glucocorticoids to mitigate inflammatory and vasculitic manifestations. Given persistent IgM-mediated disease, systemic therapy was transitioned to bendamustine combined with rituximab, a regimen associated with rapid suppression of IgM production and effective disease control in patients with Waldenström macroglobulinemia, including those with cases complicated by cryoglobulinemia.²⁴

FOLLOW-UP

Dr. Sise: Blood cultures showed clearance of *S. aureus*. The patient's condition improved, and the fever resolved. The temporary dialysis catheter was removed, and after a brief delay to allow for source control and bacterial clearance, a tunneled dialysis catheter was placed. The patient underwent five sessions of plasma exchange and

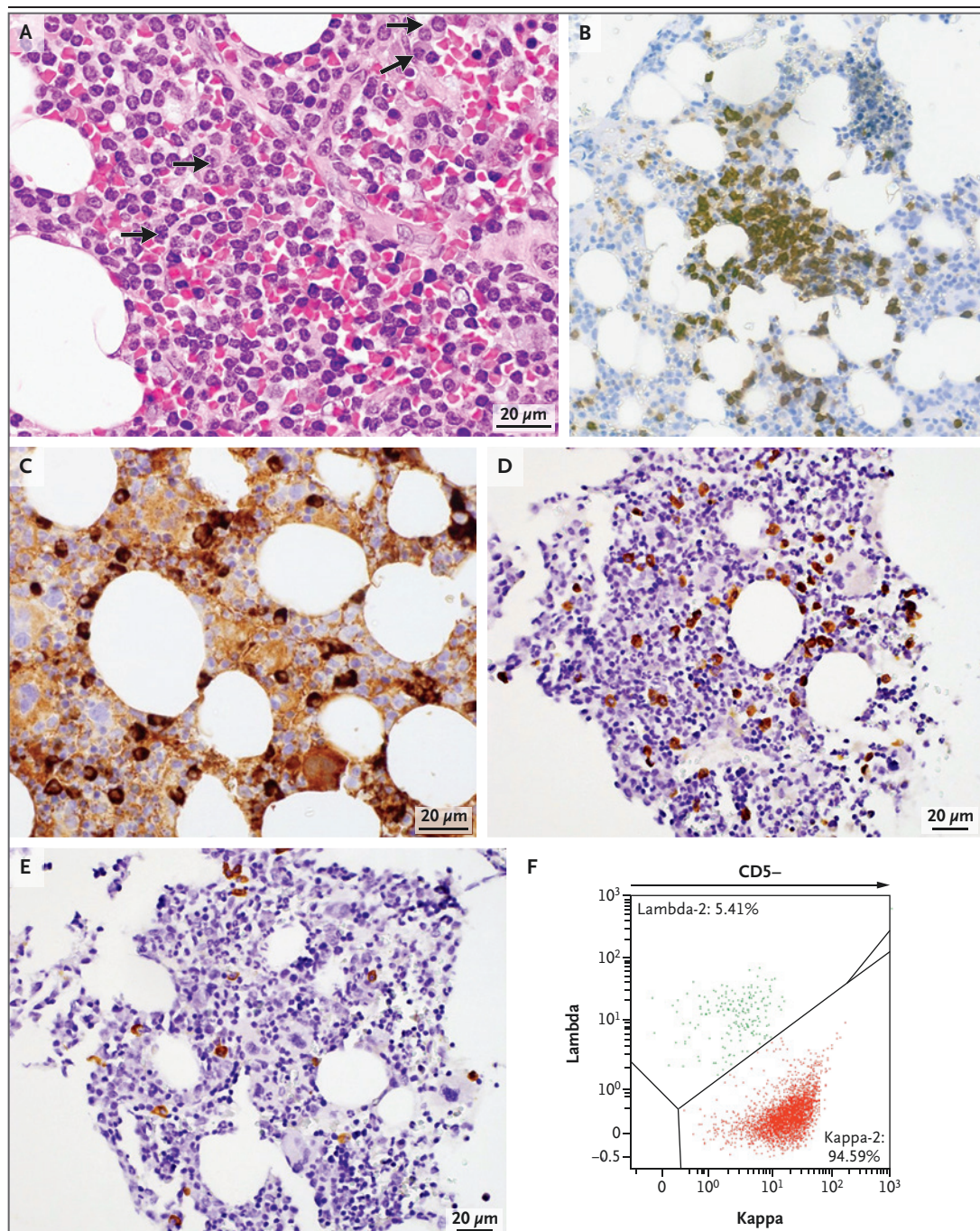


Figure 3. Bone Marrow–Biopsy Specimens and Peripheral-Blood Analysis.

Hematoxylin and eosin staining (Panel A) shows a lymphoid aggregate with scattered plasma cells (arrows). Immunohistochemical staining for CD20 (Panel B) is positive in a B-cell aggregate. Immunohistochemical staining for the μ immunoglobulin heavy chain (Panel C) shows strong IgM expression in B-cell aggregates. In situ hybridization for kappa and lambda immunoglobulin light chains (Panels D and E, respectively) shows predominant kappa light-chain expression in the plasma cells. Panel F shows representative results from flow cytometric analysis of the peripheral blood. A kappa–lambda immunoglobulin light-chain dot plot of CD5⁺ B cells shows a dominant kappa-restricted population (94.59%) and a minor lambda-restricted population (5.41%), findings consistent with a clonal B-cell process.

continued to receive hemodialysis. Transesophageal echocardiography was performed to rule out endocarditis, and a 2-week course of cefazolin was administered.

Six weeks after admission to this hospital, the patient was discharged to a rehabilitation facility, where she continued to receive hemodialysis. Her urinary output increased, and hemodialysis was discontinued 2 weeks after hospital discharge. At that time, her serum creatinine levels had improved to 1.0 to 1.2 mg per deciliter (88 to 106 μ mol per liter).

Dr. Branagan: After discharge from this hospital, the patient elected to continue oncology care at another hospital closer to her home. She continued to receive treatment with bendamustine combined with rituximab, as well as prednisone daily. Her treatment course was complicated by deep-vein thrombosis and severe hypogammaglobulinemia with recurrent infections. These complications were associated with marked functional decline, necessitating prolonged inpatient rehabilitation.

Three months after discharge from this hospital, the patient reestablished care at this hospital. The rash associated with leukocytoclastic vasculitis had resolved. Waldenström macroglobulinemia was in remission, with a very good partial response involving a reduction in IgM levels of more than 90%. Cryoglobulinemia was also in remission, with a reduction in cryoglobulin

levels from 3% of the total serum volume to less than 1%. Bendamustine combined with rituximab was discontinued after four cycles, monthly therapy with intravenous immune globulin was initiated, and the dose of glucocorticoids was tapered.

Five months after discharge from this hospital, the patient received a diagnosis of metastatic Merkel-cell carcinoma. Despite remission of Waldenström macroglobulinemia and cryoglobulinemia, the carcinoma progressed, leading to multiorgan failure. The patient was transitioned to hospice care and died peacefully at home approximately 8 weeks later.

FINAL DIAGNOSIS

Waldenström macroglobulinemia associated with type II cryoglobulinemic vasculitis.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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